Full fine-tuned neural retriever: Methodology and Evidence

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Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Test whether adapting the full in-house Phase 1 A0 language model improves case matching beyond lexical overlap. The full model has about 110.3 million trainable parameters in the saved GPU report, much more than the partial or LoRMA branches.

Architecture and mechanism

Four evenly spaced 256-token windows encode each query and candidate. The model mean-pools hidden states within each window and applies late max interaction across windows; supervised hard negatives come from the lexical retriever.

Measured result

Scored submission	1039892
Codabench micro-F1@7	0.4064
Archived local metric	0.4036 corrected local-dev micro-F1@7

Corrected local dev micro-F1@7 was 0.4036 on 118 queries; Codabench was 0.4064. The latter ties the corrected model ensemble in the screenshot.

Limitations and next check

A score tie on one hidden evaluation does not show the ensemble and full model made identical predictions. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: phase7_run.py, phase7_export.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.